



Global Income Intelligence Platform

Infosys Springboard Virtual Internship 6.0 — Completion Report

BATCH 13

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8 WEEKS · FEB-MAR 2026

Project Objective

What This Platform Sets Out to Solve

The **Global Income Intelligence Platform** is an interactive, web-based analytical system designed to study income distribution patterns and inequality trends across 200 countries. By combining exploratory data analysis, interactive dashboards, and machine learning, the platform transforms raw socioeconomic data into actionable, policy-relevant insights.



Understand Inequality

Explore global income distribution patterns and Gini index trends over time.



Interactive Dashboards

Build modular, user-friendly dashboards for dynamic data exploration.



Predictive Analytics

Integrate machine learning models to forecast future inequality trends.



Policy Insights

Enable data-driven decision making for researchers and policymakers.

Technology Stack

Tools & Frameworks Used

Core Languages & Libraries

→ **Python**

Primary programming language for all data processing and modeling tasks.

→ **Pandas**

Data ingestion, cleaning, transformation, and feature engineering.

→ **Scikit-learn**

Machine learning model building, training, and evaluation.

Visualization & Deployment

→ **Plotly & Matplotlib**

Interactive and static chart generation for in-depth visual analysis.

→ **Streamlit**

Web application framework for building the modular, browser-based dashboard.

→ **Microsoft Power BI**

Enterprise-grade BI dashboard embedded via iframe for boardroom-level reporting.

Development Timeline

8-Week Sprint: From Data to Deployment

Week 1-2

Python & Data Cleaning Loaded dataset, handled missing values with `fillna()/dropna()`, removed duplicates, and corrected data types.

1

2

Week 3

Exploratory Data Analysis Identified distribution patterns, correlations, and outliers across 3,590 rows and 20 columns of inequality data.

3

Week 4

Power BI Dashboard Built an enterprise-level visualization layer covering Gini Index trends, regional income shares, and Palma Ratio analysis.

4

Week 5-6

Streamlit App & Visualization Developed modular web interface with advanced chart explorer, country comparison, and global map visualization.

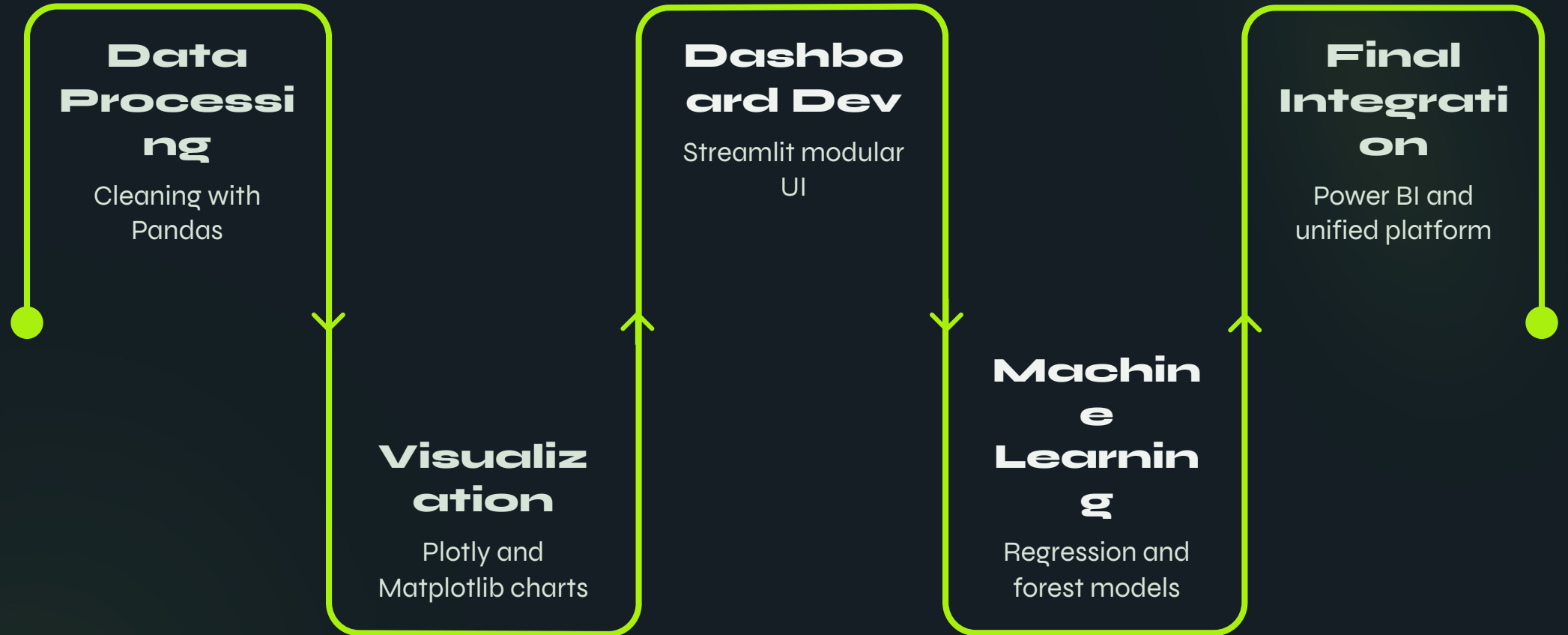
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Week 7-8

Machine Learning & Integration Trained Linear Regression, Decision Tree, and Random Forest models; integrated all components into one unified platform.

Phase-by-Phase Execution

How the Platform Was Built



The platform was developed across five tightly scoped phases. Each phase built on the last — clean data enabled richer visualizations, which in turn informed the machine learning feature selection. The final integration step brought every component under a single authenticated web interface, creating a seamless end-to-end analytics experience.

Power BI Dashboard: Global Income at a Glance

37.52

Avg Gini Index

Across all tracked countries globally.

200

Total Countries

Comprehensive worldwide coverage.

7.85bn

Population Covered

Total updated population in dataset.

22.55

Avg Inequality Index

Composite index across all regions.

The Power BI dashboard surfaces six core KPIs alongside regional breakdowns. Notable findings include Sub-Saharan Africa's Palma Ratio of 2.19 — more than **7× higher** than Europe's 0.31 — and Namibia leading the Top 5 most unequal nations with a Gini index of 64.



Power BI Dashboard: Interactive Data Explorer

Beyond static reports, the Power BI dashboard empowers users with dynamic interactivity, allowing them to dive deep into income data. Seamless navigation and customizable views facilitate comprehensive analysis, from macro trends to granular country-specific insights.

Dynamic Filters

Slice and dice data by country, region, year, and income quintile to pinpoint specific trends.

Drill-Down Capabilities

Seamlessly transition from global summaries to detailed country profiles and historical data series.

Customizable Visuals

Toggle between various chart types—bar, line, scatter—for diverse perspectives on income patterns.

Application Modules

Exploring Key Features

Beyond the Power BI dashboard, the platform offers a suite of interactive modules, built with Streamlit, designed for deeper exploration and predictive analysis of global income data.



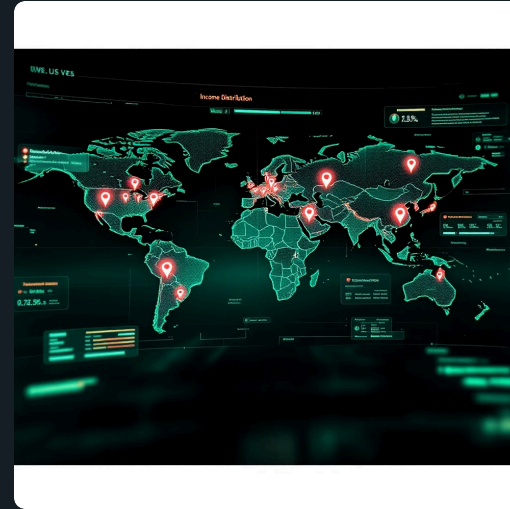
Advanced Chart Explorer

Customize and analyze income distribution metrics with a rich library of interactive charts and graphs.



Country Comparison

Directly compare economic indicators and inequality trends between multiple nations or regions.



Global Map Visualization

Gain geographical insights into income disparities and Gini index values worldwide.

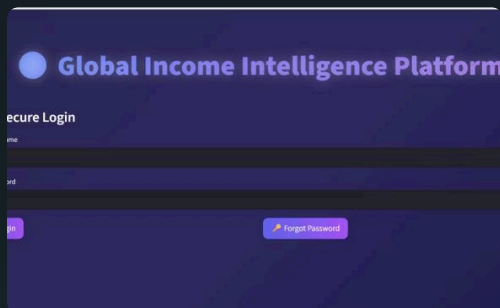


Predictive Model Interface

Utilize integrated AI/ML models to generate and evaluate future income inequality forecasts.

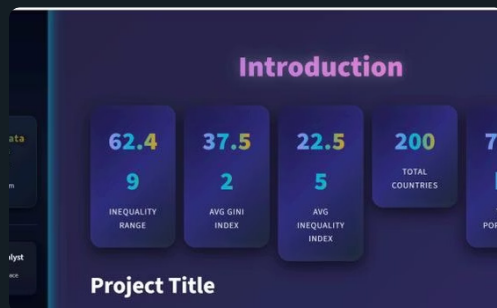
Application Modules

Platform Screenshots: Inside the Interface



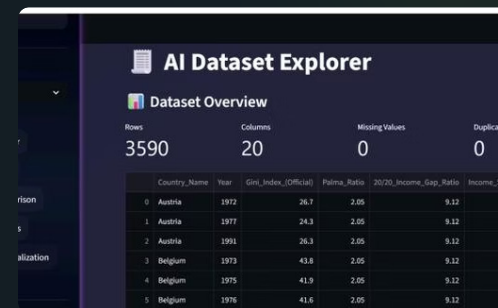
Secure Authentication

Role-based login with username/password authentication. Admin and analyst roles route users to their respective workspaces upon login.



Home Dashboard

The landing page surfaces five headline KPIs — Inequality Range (62.4), Avg Gini Index (37.5), Total Countries (200), and Total Population (7.85bn) — at a glance.



AI Dataset Explorer

Displays dataset quality metrics (3,590 rows, 20 columns, 0 missing values, 0 duplicates) and enables direct tabular exploration of cleaned inequality records by country and year.

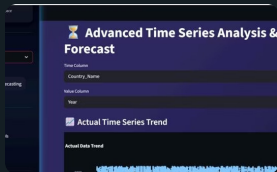
Advanced Modules

Analytics, ML & Forecasting Capabilities



ML Prediction Engine

Four models benchmarked head-to-head. **Random Forest** achieved the best R^2 score of 0.7481, far outperforming Linear Regression (0.003), validating the ensemble approach for non-linear income data.



Time Series Forecasting

The forecasting module covers 200 countries with 5 trained models. Users configure the time and value columns dynamically, generating actual trend charts and future projections for any tracked inequality metric.



The **Country Comparison Dashboard** allows analysts to select multiple countries and KPIs (e.g., Gini Index, Palma Ratio) simultaneously, rendering a side-by-side comparison table for granular cross-country analysis.

Challenges & Growth

Obstacles Overcome During Development

Data Quality Issues

The raw dataset contained missing values and inconsistent formats. Resolved through systematic preprocessing — `isnull()` identification, `fillna()/dropna()` imputation, and strict type casting using Pandas.

Streamlit UI Design

Achieving a professional, cohesive UI inside Streamlit's constraint-based layout system required custom CSS overrides and careful sidebar structuring to create a polished, enterprise-grade look and feel.

ML Model Accuracy

Initial model runs produced poor R^2 scores. Iterative feature selection, hyperparameter tuning, and switching from Linear Regression to Random Forest lifted accuracy from near-zero to 0.7481.

Deployment Configuration

Configuring the deployment environment — including dependency management, server commands, and Power BI iframe embedding — required careful debugging and platform-specific setup adjustments.

Conclusion & Acknowledgements

Over eight weeks, this internship delivered a fully functional, end-to-end analytics platform that transforms raw global income data into meaningful, policy-relevant insights. The project strengthened practical skills across data engineering, visualization, machine learning, and web deployment — forming a strong foundation for a career in data science.

Skills Acquired

- Data Analysis with Pandas
- Interactive Visualization
- ML Model Development
- Web App Deployment

Tools Mastered

- Python & Scikit-learn
- Streamlit & Plotly
- Microsoft Power BI
- Pandas & Matplotlib

Acknowledgements

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